

Can Masking Hard-to-Predict Words Improve Small Language Model Pretraining?

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Abstract

This paper asks whether small language models can pretrain more effectively when part of the masked-language-modeling budget is allocated to words that are hard for the model to predict, rather than selected uniformly at random. We study this question under BabyLM Strict-Small constraints by keeping the dataset, tokenizer, architecture, optimizer, schedule, training length, and total 15% mask rate fixed, changing only which token positions are selected for prediction. We compare several online masking signals, including repeated local token heavy hitters, recent high-loss tokens, Count-Min Sketch signals over token and character-trigram difficulty, and smoothed prediction-correctness signals. In a three-seed comparison, Heavy-Hitter Masking does not improve validation loss over random masking, although it produces small gains on the available official BabyLM diagnostics. Follow-up pilots show that raw-loss-based CMS-Morph does not improve validation loss, while a simpler correctness-smoothed Accuracy-Morph variant improves matched validation loss in two seeds and gives the clearest Entity Tracking gains. We present the results as a controlled diagnostic study rather than as a settled positive result: naive hard-token masking is insufficient, raw loss is a noisy difficulty signal, and correctness-smoothed masking is the most promising direction for future multi-seed evaluation.

1 Introduction

BabyLM-style pretraining asks whether small language models can acquire useful linguistic generalizations from developmentally plausible data budgets rather than from web-scale corpora (Warstadt et al., 2023; Hu et al., 2024; Choshen et al., 2026). In this setting, the masked-language-modeling prediction budget is especially important. A model trained with a 15% mask rate receives direct loss on only a subset of token positions, so the choice of which tokens become prediction targets may matter

more than it does in large-data pretraining.

This paper asks whether small language models can pretrain more effectively when part of the masked-language-modeling budget is allocated to words that are hard for the model to predict, rather than selected uniformly at random. Put differently: can a model learn more from the same corpus and the same number of masked tokens if some masks are spent on words it currently struggles to predict?

We study this question under a strict masking-only comparison. All systems use the same BabyLM Strict-Small data, tokenizer, masked language model architecture, optimizer, batch size, sequence length, learning-rate schedule, training length, and random seed policy. All systems also preserve the same total 15% MLM mask rate. The only intended experimental difference is mask selection: which valid token positions are chosen for prediction.

The broad idea of adaptive masking is not new. Prior work has shown that MLM masking can be improved by choosing more useful prediction targets, changing masking schedules over training, or modifying token masking probabilities according to model predictability (Gu et al., 2020; Lee et al., 2022; Yang et al., 2023; Zhong et al., 2023; Wilf et al., 2023; Edman and Fraser, 2025). Our contribution is narrower and diagnostic. We ask which online signals, if any, are useful for selecting hard-to-predict words when the model, data, training schedule, and total mask rate are fixed.

We compare several ways of operationalizing hard-to-predict masking. Heavy-Hitter Masking (HHM) uses repeated local entity-like words and recent high-loss tokens. CMS-Morph uses Count-Min Sketches over raw high-loss token strings and character trigrams, testing whether token-level difficulty should generalize across related surface forms. Accuracy-Morph replaces raw loss with smoothed top-1 prediction correctness, testing whether a candidate should become attractive only

after it has been observed enough times and is often predicted incorrectly. Entity-Accuracy-Morph tests whether adding repeated-token pressure and late adaptive decay improves the correctness-smoothed signal.

This paper contributes a controlled diagnostic study of hard-to-predict word masking for small language model pretraining. Rather than changing the data, model, tokenizer, optimizer, schedule, or total mask rate, we compare several online signals for choosing which words to mask. The results show that repeated-token and raw-loss signals are insufficient to improve validation loss in the current setup, while a correctness-smoothed masking signal gives the strongest pilot evidence and motivates future multi-seed evaluation.

2 Background and Related Work

BabyLM. BabyLM evaluates language models trained under constrained data budgets, with particular emphasis on whether small models can acquire linguistic generalizations from limited input (Warstadt et al., 2023; Hu et al., 2024; Choshen et al., 2026). The 2026 Strict-Small track limits pretraining data to 10M words or less and asks participants to implement submissions using Hugging Face Transformers. The shared-task setting is useful for this paper because it makes the central constraint explicit: performance should improve by using limited data more effectively, not by silently increasing corpus size or model scale.

The recent BabyLM literature shows that strong results often come from changing several parts of the training recipe at once. The second BabyLM findings report that the best-performing submissions tended to modify the training data, objective, or architecture, and that a hybrid causal-masked architecture was especially successful (Hu et al., 2024). GPT-BERT merges causal and masked objectives in a single transformer stack and outperforms masked-only or causal-only variants in the 2024 challenge setting (Charpentier and Samuel, 2024); AntLM similarly alternates causal and masked objectives and attention masks to combine the strengths of CLM and MLM (Yu et al., 2024). Other recent BabyLM systems explore orthogonal sources of efficiency, including linear or lightweight token mixers, optimizer changes, curated corpora, masked diffusion objectives, and frequency-informed training (Haller et al., 2025; Kosmopoulou et al., 2025). These systems are im-

portant points of comparison, but they answer a broader question than ours. We ask a narrower question: if the data, tokenizer, architecture, optimizer, schedule, training length, and total MLM mask rate are held fixed, can changing only the masking policy improve sample efficiency?

Why Mask Selection Matters. In masked language modeling, the model only receives a learning signal at the positions selected for prediction. The corpus may contain 10M words, but a 15% mask rate exposes the loss directly on only a fraction of token positions per epoch. Under web-scale pretraining this inefficiency may be tolerable because the model sees enormous redundancy. Under BabyLM Strict-Small constraints, however, the prediction budget is much tighter: repeatedly spending masks on very easy function words, isolated punctuation-adjacent tokens, or one-off examples may be less useful than spending some of the same budget on positions that stress lexical, discourse, or morphological generalization.

Adaptive and Selective Masking. Masked language modeling usually samples prediction targets uniformly from valid token positions. This is an attractive default because it is simple, unbiased with respect to token position, and does not require task-specific annotation. The downside is that random masking treats all valid positions as equally valuable prediction targets. Prior work has therefore explored masking policies that change what gets predicted. Concept-based curriculum masking evaluates token difficulty with linguistic and knowledge-graph criteria, then masks easy items first and related items later (Lee et al., 2022). Scheduled masking changes the masking ratio and masked content over training time, arguing that time-invariant MLM settings are unlikely to be optimal (Yang et al., 2023). Self-Evolution learning focuses on informative or under-explored tokens and combines adaptive masking with token-specific label smoothing (Zhong et al., 2023). Most directly related to our follow-up experiments, Edman and Fraser (2025) adapt MLM masking probabilities in BabyLM according to the model’s ability to predict tokens and also use subtoken embeddings to improve morphological generalization.

Selective and task-guided masking has also been studied outside BabyLM. Gu et al. (2020) add a task-guided pretraining stage that selectively masks task-important tokens in in-domain data.

Difference-Masking chooses masks for continued pretraining by emphasizing what distinguishes a task domain from the original pretraining domain (Wilf et al., 2023). InforMask uses pointwise mutual information to select informative tokens in an unsupervised pretraining setting (Sadeq et al., 2022). These methods support the general intuition that deciding what to mask can matter, but they often rely on task domains, downstream relevance, preprocessing statistics, or a continued-pretraining setup.

This paper does not claim to introduce adaptive masking as a new paradigm. Instead, it asks a narrower diagnostic question: under a fixed 15% masking budget and a strict masking-only comparison, which online signals are useful for selecting hard-to-predict words? We keep the total mask rate fixed for every condition and avoid adding a new objective, label smoother, subtoken embedding module, external knowledge graph, task labels, external annotations, or data-selection step. The only intended difference from the random baseline is the collator’s target-selection policy. This makes the experiment less likely to produce a large leaderboard gain, but it makes the causal interpretation cleaner: any change in validation loss or official diagnostics is attributable to which positions were predicted, not to a larger training signal.

Why Heavy Hitters. The first hypothesis behind HHM is that repeated strings in a local context are often linguistically important. In child-directed or narrative-like text, repeated names, pronouns, and content words help establish discourse participants and event structure. A random mask may hit these positions occasionally, but a heavy-hitter mechanism can deliberately spend part of the fixed mask budget on recurring entity-like mentions. This is meant to encourage the model to learn from local recurrence and to support entity-tracking behavior without adding a parser, a coreference system, or external supervision.

Why Hard Examples. The second hypothesis is curriculum-like: tokens that the current model repeatedly predicts poorly may be worth revisiting. In this paper, a hard-to-predict word is not defined by external linguistic annotation or human difficulty judgments. It is defined operationally as a token that the current model has difficulty predicting during MLM pretraining, measured either by recent high loss or by smoothed top-1 prediction

correctness. The online error component of HHM records high-loss masked targets during training and later makes matching token strings more likely to be masked. This creates a lightweight feedback loop in which the model’s own errors influence future prediction targets. The risk is that raw loss can be noisy: rare words, tokenization artifacts, or inherently ambiguous contexts may receive attention without producing useful generalization. This risk motivated the later Accuracy-Morph pilot, which replaces raw loss with smoothed prediction-correctness statistics.

Streaming Sketches. Streaming sketches are compact data structures that approximate properties of a stream, such as heavy hitters and frequency moments (Cormode and Muthukrishnan, 2005; Metwally et al., 2005). They are well matched to this setting because the masking policy observes a long stream of token events but should not store unbounded per-token histories. We use Space-Saving sketches to maintain bounded-memory summaries of repeated local entity strings and high-loss target tokens. The follow-up CMS-Morph pilots use Count-Min Sketches to track token and character-trigram difficulty signals. In both cases, the sketch is a small control mechanism over mask selection, not an added source of text or labels.

Morphological Motivation. The CMS-Morph and Accuracy-Morph pilots were tested because token-level difficulty may be too narrow in a small-data regime. If the model struggles with one word form, related forms may share prefixes, suffixes, stems, or orthographic patterns. Character trigrams are a deliberately simple proxy for these subtoken regularities: they require no morphological analyzer, work with the same tokenizer, and can generalize a difficulty signal across related surface forms. This connects to the BabyLM finding that subtoken information can improve morphological generalization (Edman and Fraser, 2025), but our implementation uses character-trigram sketches only to decide future mask positions. It does not change the input representation or add new embeddings.

Design Constraint. The central design constraint is that any observed difference should be attributable to mask selection rather than to extra data, a stronger model, a different tokenizer, or a larger prediction budget. This makes the experiment intentionally conservative. A positive result would indicate that a cheap online policy can make

the same training budget more useful; a negative or mixed result still identifies which adaptive signals are too noisy or too weak under strict controls. This is also what makes the paper different from many successful BabyLM submissions: rather than proposing a stronger complete system, we isolate one low-cost mechanism and report both where it fails and where a corrected version begins to help.

3 Hard-to-Predict Word Masking

Controlled Masking-Only Setup. All variants preserve the same 15% total MLM mask rate. They differ only in how a subset of masked positions is selected. Each collator receives tokenized examples, identifies valid non-special and non-padding positions, chooses positions to predict, applies the same BERT-style replacement policy as the random baseline, and returns labels with unmasked positions set to -100 . If an adaptive source cannot provide enough eligible positions, the remaining budget is filled with uniformly random valid positions. This fallback is part of the method because some batches simply do not contain enough repeated or previously difficult candidates.

Operational Definition. In this paper, a hard-to-predict word is not defined by external linguistic annotation. It is defined operationally as a token that the current model has difficulty predicting during MLM pretraining, measured either by recent high loss or by smoothed top-1 prediction correctness. The study compares several online signals for estimating that difficulty while keeping the model and total prediction budget fixed.

Repeated-Token Heavy-Hitter Signal. HHM operationalizes hard-to-predict masking using two signals: repeated local entity-like tokens and recent high-loss tokens. The repeated-token component is motivated by the idea that recurring names, content words, and entity-like strings often mark discourse participants or event structure. For each sequence, a lightweight heuristic extracts alphabetic non-stopword token strings that are at least two characters long and either capitalized or content-word-like. A bounded Space-Saving sketch counts candidates locally, and positions belonging to repeated top strings are eligible for the entity-heavy-hitter share. This component does not use a parser, coreference system, named-entity recognizer, or external supervision.

Raw-Loss Difficulty Signal. The second HHM signal uses the model’s own recent prediction failures. After warmup, the training loop computes per-token MLM losses for masked positions and updates a bounded Space-Saving sketch with the top-loss target token strings in each batch. Later batches preferentially mask positions whose token string appears in this error sketch. The default HHM mixture masks 15% of valid positions, with intended shares of 50% random, 25% repeated-token heavy hitters, and 25% raw-loss heavy hitters. This tests the most direct form of hard-to-predict masking: revisit words that recently produced high loss.

Token and Character-Trigram Difficulty Sketches. CMS-Morph tests whether raw-loss difficulty should generalize beyond exact token strings. It maintains Count-Min Sketches over high-loss token strings and character trigram features extracted from high-loss tokens. Character trigrams are a deliberately simple proxy for subtoken regularities: if related forms share suffixes, prefixes, stems, or orthographic patterns, a trigram sketch can transfer difficulty information across forms without changing the tokenizer or adding a morphological analyzer. CMS-Morph preserves the 15% mask rate but allocates masks after warmup to 70% random, 20% token-sketch-guided, and 10% trigram-sketch-guided positions.

Correctness-Smoothed Difficulty Signal. Accuracy-Morph tests whether raw loss is too noisy and whether smoothed prediction correctness gives a cleaner difficulty estimate. For every masked token during training, the model’s top-1 prediction is compared with the true MLM label. Token sketches count masked-token exposures and wrong predictions; analogous character-trigram sketches count feature exposures and wrong predictions. Candidate positions are scored by smoothed error rate,

$$\frac{\text{wrong} + 0.5}{\text{seen} + 1.0},$$

so a token or trigram becomes attractive only after it has been observed enough times and is often predicted incorrectly. Accuracy-Morph uses a longer 30% warmup and a conservative 80/15/5 random/token/trigram mask mixture, again preserving the 15% total mask rate.

Entity-Accuracy-Morph. Entity-Accuracy-Morph tests whether the correctness-smoothed

signal benefits from adding repeated-token pressure and late adaptive decay. It uses the same correctness sketches as Accuracy-Morph, adds a small bonus for repeated non-special token strings within a batch, filters adaptive candidates to an intermediate predicted-error band, and decays the adaptive share late in training. This variant asks whether the repeated-token intuition from HHM can be combined with the stronger correctness-smoothed signal without changing the model or prediction budget.

Design Principle. The method deliberately avoids changing the model, tokenizer, dataset, optimizer, schedule, training length, or total mask rate. This keeps the comparison focused on the masking signal rather than on capacity, compute, or additional supervision.

4 Experimental Setup

Data. We use the official detoxified BabyLM 2026 Strict-Small training set released by the BabyLM community. The dataset contains 10M words drawn from BNC spoken, CHILDES, Gutenberg, OpenSubtitles, Simple Wikipedia, and Switchboard sources. All compared systems use the same train/validation split.

Systems. The main three-seed comparison is between a standard random masked language modeling baseline and HHM, the first hard-to-predict masking operationalization. We additionally implement ablations for entity-only masking, error-only masking, a randomized sketch control, and a frequency-weighted masking control. After the main comparison, we ran adaptive masking pilots that test alternative difficulty signals: CMS-Morph replaces HHM’s entity/error sources with Count-Min Sketch scores over high-loss token strings and high-loss character trigram features; Accuracy-Morph uses smoothed top-1 prediction correctness instead of raw loss; and Entity-Accuracy-Morph adds repeated-token scoring and late adaptive decay on top of Accuracy-Morph. CMS-Morph and Entity-Accuracy-Morph were run for seed 1; Accuracy-Morph was run for seeds 1 and 2 after the seed-1 pilot looked promising.

Why These Pilots. The follow-up pilots were chosen to test specific explanations for the HHM outcome rather than to tune toward the validation set. CMS-Morph asks whether the original token-level hard-example signal was too nar-

row and should generalize through subtoken patterns. Accuracy-Morph asks whether raw loss was too noisy and should be replaced by a more stable correctness rate. Entity-Accuracy-Morph asks whether the best pilot signal can be combined with repeated-token/entity pressure and a late return toward random masking. Each pilot keeps the same model and total mask rate, so the comparison remains about mask selection rather than capacity or compute.

The pilots also position the paper relative to prior adaptive masking work. Concept-based curriculum masking and scheduled masking show that changing what gets masked over training can improve MLM efficiency (Lee et al., 2022; Yang et al., 2023). BabyLM adaptive MLM work shows that model-predictability-aware masking and subtoken information can help in the strict-small setting (Edman and Fraser, 2025). Our pilots ask a smaller mechanistic question under tighter controls: which online signal is least noisy when the only allowed intervention is reweighting a fixed 15% mask budget? In our runs, raw cross-entropy loss and an added entity bonus are not enough; smoothed top-1 correctness with a small character-trigram component is the most promising signal.

Model and Training. All systems use the same Hugging Face Transformers masked language model, tokenizer, optimizer, batch size, sequence length, learning-rate schedule, random seed policy, training steps, and total mask rate. The only intended experimental difference is the masking policy. This differs from stronger BabyLM system submissions such as GPT-BERT, AntLM, BLaLM, or masked diffusion models, where the training objective, architecture, optimizer, or model family changes (Charpentier and Samuel, 2024; Yu et al., 2024; Haller et al., 2025; Kosmopoulou et al., 2025). Those approaches are better candidates for maximizing leaderboard performance; our setup is better suited for isolating whether target selection alone carries useful signal.

Evaluation. We report validation MLM loss from the local training pipeline after 10k training steps. This is the closest measure to the training objective: the model is evaluated on held-out BabyLM-style text with the same masked language modeling loss, and lower loss indicates better average prediction of masked validation tokens. Validation loss is therefore the cleanest measure of

sample-efficient pretraining under the objective we optimize, but it does not directly test whether the model has acquired specific linguistic generalizations.

We also evaluate checkpoints with the official BabyLM 2026 strict zero-shot pipeline for the tasks available in the local evaluation bundle. Table 1 summarizes the reported benchmarks. These evaluations use the trained masked language model without task-specific fine-tuning. For the accuracy-style benchmarks, the scorer compares the model’s preference over controlled alternatives, such as a grammatical versus ungrammatical sentence or a correct versus incorrect continuation. For reading, the scorer compares model-derived surprisal with human reading measurements and reports separate eye-tracking and self-paced reading scores. Higher is better for all official metrics.

BLiMP. BLiMP is a broad zero-shot linguistic acceptability suite built from minimal pairs. Each item contrasts a sentence that satisfies a targeted linguistic constraint with a minimally different sentence that violates it. The local report groups BLiMP items into fields such as morphology, semantics, syntax, and syntax/semantics, with individual phenomena including determiner-noun agreement, subject-verb agreement, binding, island constraints, negative polarity item licensing, passives, argument structure, and wh-dependencies. A model receives credit when it assigns the better score to the acceptable member of the pair.

BLiMP Supplement. The BLiMP Supplement extends the minimal-pair style evaluation to additional targeted phenomena not covered by the main BLiMP suite. In the local bundle, the reported subsets include hypernymy, subject-auxiliary inversion, easy and tricky question-answer congruence, and turn taking. This metric is useful because it tests a mixture of syntactic, semantic, and discourse-pragmatic contrasts that may not be captured by the main BLiMP categories.

Entity Tracking. Entity Tracking is especially relevant to HHM because the method explicitly masks repeated entity-like strings. The benchmark tests whether a model can maintain information about discourse entities across controlled contexts. The local report includes regular cases, ambiguous-reference cases, and move-contents cases, each evaluated across increasing numbers of operations. Higher scores indicate that the model more often

prefers the continuation or alternative consistent with the correct entity state. Because HHM was designed partly around discourse recurrence, Entity Tracking is the most directly motivated official diagnostic for the entity-heavy-hitter component.

COMPS. COMPS evaluates compositional generalization. The local report includes a base condition and nonce or “wug”-style conditions, including variants with distractors before or between relevant material. These examples test whether the model can combine lexical and structural information productively rather than relying only on memorized surface strings. This benchmark is relevant to the morph pilots because character-trigram adaptive masking is intended to encourage some generalization across related or novel-looking forms.

Reading. The reading benchmark evaluates whether the model’s word-by-word expectations align with human processing data. Instead of minimal-pair accuracy, the official report provides separate eye-tracking and self-paced reading scores. Intuitively, a better language model should assign higher surprisal to words that humans also find harder or slower to process, although the relationship is indirect. We treat these scores as psycholinguistic diagnostics rather than as direct measures of MLM training quality.

The main baseline and HHM checkpoints are evaluated across three seeds with both local validation loss and the available official metrics. The CMS-Morph and Entity-Accuracy-Morph pilots are evaluated with the same local and official commands for one seed, while Accuracy-Morph is evaluated for two matched seeds. We report these follow-ups separately from the main three-seed comparison because they were selected adaptively after the initial HHM result. EWoK was not present in the local evaluation data, so it is not reported.

Diagnostics. Each run logs the number of total masked tokens and the number selected from random, repeated-token, raw-loss, token-accuracy, or morph-accuracy sources as applicable. This verifies that each adaptive policy reallocates the prediction budget without increasing it.

5 Results

Table 2 reports validation loss for the completed 10k-step main comparison, along with official BabyLM 2026 strict zero-shot evaluation results

Metric	Evaluation type	What it probes	Reported score
Validation MLM loss	In-domain held-out MLM	Fit to the held-out BabyLM-style validation split under the training objective	Mean cross-entropy; lower is better
BLiMP	Zero-shot minimal-pair preference	Morphology, syntax, syntax/semantics, negative polarity items, agreement, islands, binding, argument structure	Average accuracy; higher is better
BLiMP Supplement	Zero-shot minimal-pair preference	Additional targeted phenomena including hypernymy, question-answer congruence, subject-auxiliary inversion, and turn taking	Average accuracy; higher is better
Entity Tracking	Zero-shot controlled reference task	Whether the model tracks discourse entities through regular, ambiguous-reference, and move-contents conditions across increasing operation counts	Average accuracy; higher is better
COMPS	Zero-shot compositional generalization	Compositional interpretation, including nonce/wug-style conditions and distractor variants	Average accuracy; higher is better
Reading	Psycholinguistic prediction	Whether model surprisal aligns with human reading behavior in eye-tracking and self-paced reading data	Official reading scores; higher is better

Table 1: Evaluation metrics reported in this paper. The official BabyLM strict zero-shot pipeline scores the available benchmark files in the local evaluation bundle. Accuracy-style tasks compare model preferences over controlled alternatives; reading evaluates alignment between model-derived surprisal and human reading measurements.

for the tasks available in the local evaluation bundle. EWok was not present in the local evaluation data, so we omit it rather than substituting a non-identical task source.

Does Hard-to-Predict Masking Improve Validation Loss? The main comparison is Baseline Random MLM versus HHM Entity+Error, with three 10k-step seeds each. The random baseline reaches a mean validation loss of 3.0023 ± 0.1576 , while HHM reaches 3.0321 ± 0.1469 , a $+0.0298$ increase in loss. On local MLM validation loss, HHM is therefore slightly worse than random masking in the completed matched-run comparison, although the gap is small relative to seed-to-seed variation.

Does It Affect Official BabyLM Diagnostics? The official zero-shot results point in the opposite direction, but only weakly. As summarized in Table 3, HHM is higher than the random baseline on every reported official metric: BLiMP improves by $+0.23$, BLiMP Supplement by $+0.11$, Entity Tracking by $+0.88$, COMPS by $+0.37$, eye-tracking reading score by $+0.31$, and self-paced reading score by $+0.50$. These are small gains, and Entity Tracking in particular has high HHM variance (± 4.39), so the official evaluation should be read as suggestive rather than decisive.

This pattern is important for interpretation. If we judged only by official averages, HHM would look mildly encouraging; if we judged only by validation loss, it would look negative. The controlled design makes the tension visible. HHM changes the prediction targets but not the prediction budget, and the result is not a general sample-efficiency improvement. Instead, it appears to move some linguistic diagnostics without improving the objective used for pretraining. This is weaker than the gains reported by broader BabyLM systems that change architecture or objective, but it is a cleaner test of mask selection itself.

Are the Main Differences Statistically Established? Table 4 reports paired tests over the three matched seeds. None of the observed differences

are statistically significant. The validation-loss gap against HHM has paired t -test $p = 0.114$ and exact sign-flip $p = 0.25$; COMPS and self-paced reading also have exact sign-flip $p = 0.25$ because all three paired deltas go in the same direction, but with only three seeds this is still the strongest possible two-sided exact result. We therefore interpret all main HHM differences as descriptive trends, not as statistically established effects.

What Is the Bottom Line for HHM? The official evaluation answers the main no-retraining question: the existing HHM checkpoints do show modest behavioral improvements despite slightly worse MLM validation loss. This weakens a purely negative interpretation of HHM, but does not establish a robust sample-efficiency gain. The bottom line is mixed: HHM is worse on the training objective used for selection, while the available official diagnostics show small average improvements.

Are Raw-Loss Signals Useful? Table 5 reports three additional controlled seed-1 pilots. CMS-Morph keeps the same model, tokenizer, data, optimizer, schedule, training length, and 15% total mask rate, but after a 10% warmup it allocates 70% of masks randomly, 20% using a Count-Min Sketch over high-loss token strings, and 10% using a Count-Min Sketch over high-loss character trigrams. The realized final batches used the intended fixed budget: 1,216 masked tokens per batch, split into 832 random and 384 adaptive token/trigram-guided masks.

CMS-Morph is not a clear improvement. Its final validation loss is 3.1574, slightly worse than the matched seed-1 random baseline at 3.1445 and slightly worse than HHM seed 1 at 3.1530. On official metrics, however, it is mixed rather than uniformly negative: relative to random seed 1, CMS-Morph improves BLiMP by $+0.52$, COMPS by $+0.23$, eye-tracking by $+1.72$, and self-paced reading by $+0.52$, while decreasing BLiMP Supplement by -1.36 and Entity Tracking by -0.03 . We therefore treat CMS-Morph as a useful additional probe, not as a replacement result: subtoken sketch

System	Seeds	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM baseline	3	3.0023 $\pm$ 0.1576	62.43 $\pm$ 0.62	55.77 $\pm$ 0.16	16.77 $\pm$ 0.93	50.18 $\pm$ 0.30	1.97 $\pm$ 0.92	2.14 $\pm$ 0.38
HHM entity+error	3	3.0321 $\pm$ 0.1469	62.66 $\pm$ 0.51	55.88 $\pm$ 1.78	17.65 $\pm$ 4.39	50.55 $\pm$ 0.47	2.28 $\pm$ 0.14	2.64 $\pm$ 0.25

Table 2: Main results. Values are mean $\pm$ sample standard deviation over three 10k-step seeds. Validation loss is local MLM loss, where lower is better. Evaluation columns are official BabyLM strict zero-shot outputs available in the local evaluation bundle; higher is better. The local bundle did not include EWoK data, so EWoK is omitted.

Metric	Baseline	HHM	Δ
MLM validation loss	3.0023	3.0321	+0.0298
BLiMP	62.43	62.66	+0.23
BLiMP Supplement	55.77	55.88	+0.11
Entity Tracking	16.77	17.65	+0.88
COMPS	50.18	50.55	+0.37
Reading: eye tracking	1.97	2.28	+0.31
Reading: self-paced	2.14	2.64	+0.50

Table 3: Mean HHM change relative to the random masking baseline. Lower is better for MLM validation loss, so the positive loss delta is worse; higher is better for all official evaluation metrics.

guidance can move some official diagnostics, but raw-loss token and trigram sketches do not improve the core training objective in this setup.

Is Smoothed Correctness a Better Signal?

Accuracy-Morph is more promising in the seed-1 pilot. It uses the same fixed 15% mask rate, but after a 30% warmup allocates 80% of masks randomly, 15% by token correctness sketches, and 5% by character trigram correctness sketches. The final batches again preserve the fixed budget, with 960 random and 256 adaptive masks per batch. Relative to random seed 1, Accuracy-Morph lowers validation loss by -0.0119 and improves Entity Tracking by $+3.20$, COMPS by $+0.36$, eye-tracking by $+0.55$, and self-paced reading by $+0.33$, while slightly decreasing BLiMP by -0.15 and BLiMP Supplement by -0.64 . This suggested that correctness-smoothed adaptive masking was a better next direction than raw loss-guided CMS-Morph, motivating the seed-2 replication below.

The final limited-compute run replicates Accuracy-Morph with seed 2. As shown in Table 6, Accuracy-Morph seed 2 improves matched validation loss from 3.0296 to 3.0105, a -0.0191 change. It also improves Entity Tracking by $+7.82$, eye-tracking by $+0.21$, self-paced reading by $+0.04$, and BLiMP Supplement by $+0.12$, while leaving BLiMP unchanged and decreasing COMPS by -0.37 . This is the strongest current evidence for the masking idea: the raw-loss and entity-bonus variants remain negative or mixed, but correctness-

smoothed masking now improves matched validation loss in an additional seed while preserving the fixed 15% mask budget.

Table 7 summarizes the matched two-seed comparison. Accuracy-Morph improves validation loss in both seed 1 and seed 2, with a mean paired validation-loss change of -0.0155 ± 0.0051 . Its largest diagnostic effect is Entity Tracking, where the mean paired gain is $+5.51 \pm 3.27$. The rest of the official metrics are mixed: BLiMP is nearly unchanged, Supplement is lower on average because of seed 1, COMPS is effectively flat, and both reading scores improve modestly. Table 8 gives the broader context, comparing the two Accuracy-Morph seeds against all three available random seeds. This is promising pilot evidence, not a settled result: Accuracy-Morph still needs a broader multi-seed evaluation before it can support a strong positive claim.

The Accuracy-Morph result also clarifies what the paper can and cannot claim. It does not show that any adaptive mask is useful: CMS-Morph uses a larger adaptive share and still hurts validation loss. It also does not show that entity-like recurrence is sufficient: the entity-augmented variant loses the Accuracy-Morph advantage. The evidence instead points to a specific design lesson that matches the literature on model-aware masking: the adaptive signal should be stable and calibrated to the model’s actual prediction behavior. In this setting, smoothed correctness is more reliable than raw loss.

Entity-Accuracy-Morph tests a more targeted variant of Accuracy-Morph by adding repeated-token scoring, filtering adaptive candidates to an intermediate predicted-error band, and decaying the adaptive share late in training. This did not improve the result. The best validation checkpoint reached 3.1342 at step 8,000, close to Accuracy-Morph, but the final validation loss rose to 3.1566. Relative to random seed 1, the official metrics improve Entity Tracking by $+2.01$, eye-tracking by $+0.53$, and self-paced reading by $+0.28$, but BLiMP falls by -0.68 and COMPS by -0.06 . The final batches

Metric	Baseline mean	HHM mean	Δ HHM-Base	Paired t p	Exact sign-flip p
MLM validation loss	3.0023	3.0321	+0.0297	0.114	0.25
BLiMP	62.43	62.66	+0.23	0.757	1.00
BLiMP Supplement	55.77	55.88	+0.11	0.929	1.00
Entity Tracking	16.77	17.65	+0.88	0.721	0.75
COMPS	50.18	50.55	+0.38	0.177	0.25
Reading: eye tracking	1.97	2.28	+0.32	0.559	0.75
Reading: self-paced	2.14	2.64	+0.50	0.166	0.25

Table 4: Paired significance tests over the three matched random-vs-HHM seeds. Positive validation-loss deltas are worse because lower MLM loss is better; positive official-metric deltas are better. With only three paired seeds, the tests are intentionally low-power: the smallest possible two-sided exact sign-flip p value is 0.25.

System	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM, seed 1	3.1445	62.81	55.81	15.84	50.10	0.98	1.78
HHM, seed 1	3.1530	62.41	53.84	15.90	50.82	2.16	2.43
CMS-Morph, seed 1	3.1574	63.33	54.45	15.81	50.33	2.70	2.30
Accuracy-Morph, seed 1	3.1326	62.66	55.17	19.04	50.46	1.53	2.11
Entity-Accuracy-Morph, seed 1	3.1566	62.13	55.17	17.85	50.04	1.51	2.06
CMS-Morph - Random	+0.0129	+0.52	-1.36	-0.03	+0.23	+1.72	+0.52
Accuracy-Morph - Random	-0.0119	-0.15	-0.64	+3.20	+0.36	+0.55	+0.33
Entity-Accuracy-Morph - Random	+0.0121	-0.68	-0.64	+2.01	-0.06	+0.53	+0.28

Table 5: Additional one-seed adaptive masking pilots. CMS-Morph uses high-loss token and character trigram Count-Min Sketches; Accuracy-Morph instead uses smoothed prediction-correctness sketches and a more conservative adaptive share; Entity-Accuracy-Morph adds a repeated-token/entity bonus and late adaptive decay. All keep the total MLM mask rate fixed at 15%. Lower is better for validation loss; higher is better for official metrics.

preserved the fixed mask budget, with 1,152 random masks and only 64 adaptive masks after late decay. We therefore treat this run as evidence against adding entity bonuses and late adaptive decay in the current form; the cleaner Accuracy-Morph pilot remains the stronger follow-up.

Table 9 summarizes the current situation. The paper should not claim that HHM is a strong positive result: the three-seed validation-loss comparison is negative, while the official metrics provide only suggestive behavioral gains. Among the follow-up pilots, Accuracy-Morph is now the only branch with a replicated matched-seed validation-loss improvement and a strong entity-tracking diagnostic. The recommended continuation is therefore an additional Accuracy-Morph seed, not raw-loss or entity-bonus variants.

Ablation Results. Table 10 reports single-seed preliminary ablations for entity-only masking, error-only masking, randomized sketch masking, and frequency-only masking. These runs were short pipeline checks rather than full comparisons: their validation losses remain near 8.65–8.72 after 20 steps, with the randomized sketch control

lowest among them at 8.6462. They are useful for verifying implementation coverage but should not be interpreted as substantive ablation evidence. The current completed evidence is therefore the three-seed random-vs-HHM comparison plus the adaptive follow-up runs, not a full multi-seed ablation suite.

What Do the Masking Diagnostics Show? Table 11 summarizes the HHM mask-source logs. The observed HHM allocation is 15.4% entity-selected masks, 18.9% error-selected masks, and 65.6% random masks. This confirms that the collator is reallocating part of the fixed 15% mask budget toward heavy hitters, while also showing that fallback to random masking is common under the current heuristics. Table 12 reports the corresponding final-step budget checks for the adaptive pilots.

6 Masking Analysis

We analyze whether HHM changes the distribution of prediction targets while preserving the same total number of masked tokens as the baseline. The implementation logs total masked tokens, entity-selected masks, error-selected masks, and fallback random masks for every training step.

Fairness Checks. Unit tests verify that random masking and HHM mask the same number of valid positions, that padding and special tokens are never masked, that entity positions correspond to repeated local strings, that the error sketch is inactive during warmup, and that masking is deterministic under a fixed seed.

Observed Allocation. Across the three completed 10k-step HHM runs, the collator masks the same average number of tokens per step as the random baseline: 1215.8 valid tokens. The source of

System	Seed	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM	2	3.0296	62.76	55.91	17.70	50.51	2.12	2.53
Accuracy-Morph	2	3.0105	62.76	56.03	25.52	50.14	2.33	2.57
Accuracy-Morph – Random	2	-0.0191	+0.00	+0.12	+7.82	-0.37	+0.21	+0.04

Table 6: Final limited-compute experiment: Accuracy-Morph seed 2 compared with the matched random MLM seed 2 baseline. Lower is better for validation loss; higher is better for official metrics.

System	Seeds	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM	1,2	3.0871 $\pm$ 0.0812	62.79 $\pm$ 0.04	55.86 $\pm$ 0.07	16.77 $\pm$ 1.32	50.31 $\pm$ 0.29	1.55 $\pm$ 0.81	2.16 $\pm$ 0.53
Accuracy-Morph	1,2	3.0715 $\pm$ 0.0863	62.71 $\pm$ 0.07	55.60 $\pm$ 0.61	22.28 $\pm$ 4.58	50.30 $\pm$ 0.23	1.93 $\pm$ 0.57	2.34 $\pm$ 0.33
Paired delta	1,2	-0.0155 $\pm$ 0.0051	-0.08 $\pm$ 0.11	-0.26 $\pm$ 0.54	+5.51 $\pm$ 3.27	-0.00 $\pm$ 0.52	+0.38 $\pm$ 0.24	+0.19 $\pm$ 0.21

Table 7: Matched-seed Accuracy-Morph comparison. The paired delta is Accuracy-Morph minus Random MLM for seeds 1 and 2. Accuracy-Morph improves validation loss in both matched seeds and produces a large positive Entity Tracking delta, while BLiMP, Supplement, and COMPS remain roughly flat or slightly lower.

those HHM masks is 15.4% entity-selected positions, 18.9% error-selected positions, and 65.6% fallback random positions. This verifies that HHM changes the target distribution without increasing the total prediction budget, but it also shows that the intended 25/25/50 entity-error-random mixture is not fully realized.

Interpretation. The high fallback rate gives a concrete explanation for the weak validation-loss result. Entity and error sketches are active, but the realized intervention is closer to a mostly random masking policy with a moderate heavy-hitter bias than to an evenly balanced three-source mixture. The official evaluation suggests that this moderate shift may still affect measured linguistic behavior: HHM is slightly higher than the random baseline on the available official zero-shot tasks, including Entity Tracking and COMPS. However, the gains are small and often well within seed variation, so they should be treated as a diagnostic signal rather than a reliable win.

The pilot diagnostics point to a narrower lesson. Increasing adaptivity is not automatically helpful: CMS-Morph assigns a larger final adaptive share than Accuracy-Morph but has worse validation loss, while Entity-Accuracy-Morph decays to a very small final adaptive share and also loses the Accuracy-Morph advantage. The useful signal appears to be the scoring rule, not simply the amount of non-random masking. In the current runs, smoothed prediction correctness is a cleaner adaptive signal than raw loss or repeated-token bonuses.

7 Discussion

The results do not support naive hard-to-predict word masking as a reliable improvement over random masking. HHM is mechanically viable and respects the fairness constraint, but it does not improve validation loss in the completed three-seed comparison. Raw-loss CMS-Morph also fails to improve validation loss, even though it assigns a larger adaptive share than Accuracy-Morph. The negative result is therefore not simply that the adaptive share was too small; the difficulty signal itself matters.

The official BabyLM diagnostics make the result more nuanced. HHM is slightly higher than the random baseline on each available official zero-shot metric, even though its validation loss is worse. These gains are small and not statistically established, but they show that hard-to-predict masking can alter measured linguistic behavior without improving the local MLM objective. This is a useful diagnostic outcome: validation loss and downstream-style BabyLM diagnostics need not move in lockstep under small masking-policy changes.

The strongest lesson is that raw high loss is a noisy online difficulty signal. A high-loss token may be rare, ambiguous, poorly tokenized, or unlearnable in context. Reallocating masks toward such tokens does not necessarily create better pretraining signal. Smoothed prediction correctness appears more stable in the current experiments because it estimates repeated prediction failure over exposures rather than reacting directly to loss magnitude. Accuracy-Morph is therefore the most promising tested signal: across

System	Seeds	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM	1,2,3	3.0023 $\pm$ 0.1576	62.43 $\pm$ 0.62	55.77 $\pm$ 0.16	16.77 $\pm$ 0.93	50.18 $\pm$ 0.30	1.97 $\pm$ 0.92	2.14 $\pm$ 0.38
Accuracy-Morph	1,2	3.0715 $\pm$ 0.0863	62.71 $\pm$ 0.07	55.60 $\pm$ 0.61	22.28 $\pm$ 4.58	50.30 $\pm$ 0.23	1.93 $\pm$ 0.57	2.34 $\pm$ 0.33

Table 8: Aggregate view after the final Accuracy-Morph seed-2 run. Accuracy-Morph now has two seeds, while the random baseline has three. The aggregate is therefore useful as context but not a perfectly balanced significance test.

Branch	Evidence	Seeds	Current interpretation
HHM entity+error	Worse MLM loss than random (+0.0298), but small gains on all available official metrics	3	Mixed diagnostic result; not a clear training-objective improvement
CMS-Morph	Worse seed-1 MLM loss than random (+0.0129), mixed official metrics	1	Useful probe, but raw high-loss sketches are not the best next path
Accuracy-Morph	Better matched seed-2 MLM loss than random (-0.0191), unchanged BLiMP, and much higher Entity Tracking (+7.82)	2	Best current follow-up; positive but still needs a third seed
Entity-Accuracy-Morph	Worse seed-1 MLM loss than random (+0.0121), smaller Entity Tracking gain than Accuracy-Morph	1	Do not continue this variant as the main direction

Table 9: Current status of the experimental branches. Positive validation-loss deltas are worse because lower MLM loss is better; official metric deltas are interpreted only as diagnostics unless replicated across seeds.

Condition	Steps	Validation loss
Random baseline	20	8.6524
HHM entity+error	20	8.6703
Entity only	20	8.6535
Error only	20	8.6611
Random sketch control	20	8.6462
Frequency only	20	8.7238

Table 10: Preliminary single-seed component ablations. These are 20-step pipeline checks rather than full training runs, so they verify implementation behavior but are not used as substantive evidence about final model quality.

System	Entity	Error	Random
HHM entity+error	15.4%	18.9%	65.6%

Table 11: Observed source of masked positions in the HHM diagnostic logs, aggregated over 30,000 logged batches from the three 10k-step HHM seeds. The target mixture is 25% entity, 25% error, and 50% random, with random fallback used when a heavy-hitter source lacks enough valid positions.

seeds 1 and 2 it improves matched validation loss by -0.0155 ± 0.0051 and Entity Tracking by $+5.51 \pm 3.27$. This is encouraging pilot evidence, not a confirmed positive result.

The entity-augmented follow-up is also instructive. Adding repeated-token pressure, an intermediate difficulty band, and late adaptive decay does not preserve the Accuracy-Morph advantage. This suggests that future work should not simply add more heuristics. The most defensible next step is a multi-seed Accuracy-Morph evaluation with cleaner prediction-correctness scoring, modest token and morph adaptive shares, and carefully tested difficulty-band filtering.

Relative to the closest ACL Anthology work, the main difference is experimental scope. Prior

Pilot	Final masks	Random masks	Adaptive masks
CMS-Morph	1,216	832	384
Accuracy-Morph	1,216	960	256
Entity-Accuracy-Morph	1,216	1,152	64

Table 12: Final-step mask-budget diagnostics for the seed-1 adaptive pilots. All pilots preserve the same 15% total MLM mask rate; they differ only in how much of that fixed budget is assigned to adaptive token or character-trigram sources. Entity-Accuracy-Morph decays to a much smaller adaptive share late in training.

adaptive-masking papers show that changing masked content, schedules, task-guided selection, or predictability-aware token probabilities can improve pretraining or BabyLM performance (Gu et al., 2020; Lee et al., 2022; Yang et al., 2023; Wilf et al., 2023; Edman and Fraser, 2025). BabyLM system papers show that larger changes, such as hybrid causal-masked objectives, lightweight architectures, curated data, optimizer changes, or masked diffusion, can be stronger overall routes to leaderboard performance (Hu et al., 2024; Charpentier and Samuel, 2024; Yu et al., 2024; Haller et al., 2025; Kosmopoulou et al., 2025). Our experiments are deliberately narrower. The value of the paper is not that HHM beats those systems; it does not. The value is that it isolates the masking-policy variable and documents which online signals survive that constraint. Under this stricter test, repeated-token and raw-loss signals are insufficient, while correctness-smoothed token and trigram sketches are a plausible low-cost direction.

8 Limitations

The broad idea of adaptive or selective masking has substantial prior work. This paper does not claim to introduce adaptive masking, prediction-guided masking, or selective masking as new paradigms. Its contribution is the narrower controlled com-

parison of several online hard-to-predict masking signals under the same small-language-model pre-training setup.

The main HHM result does not improve validation loss. The available official BabyLM diagnostic gains are small, descriptive, and not statistically significant in the three-seed paired tests. They should be treated as suggestive diagnostics rather than as evidence of a robust downstream improvement.

Accuracy-Morph is the most promising follow-up, but it has only two matched seeds. It should not be presented as a confirmed method. The safe conclusion is that smoothed prediction correctness deserves multi-seed follow-up, not that hard-to-predict word masking has been shown to improve small language model pretraining.

The term hard-to-predict is operationalized by model prediction behavior during MLM pretraining, not by human judgments of linguistic difficulty or external linguistic annotation. A token can be hard for many reasons, including rarity, ambiguity, tokenization, or insufficient context. This is one reason raw high-loss signals can be noisy.

The current implementation uses heuristic token-level entity extraction rather than a full syntactic or coreference system. Error heavy hitters are keyed by target token string in the first implementation, so they do not distinguish different contexts for the same token. The local BabyLM evaluation bundle did not include EWoK, so we do not report EWoK and cannot assess that part of the full BabyLM suite here. The component ablations reported here are short 20-step checks and should not be read as final ablation evidence.

The paper also does not attempt to match the strongest BabyLM systems. We do not use hybrid causal-masked objectives, architecture changes, optimizer changes, curated corpora, external lexical resources, downstream task labels, or auxiliary representation modules. That restriction is scientifically useful for attribution, but it limits expected absolute performance and means the conclusions depend on this specific small LM architecture and 10k-step training budget. A stronger final system would likely combine the best controlled masking signal here, Accuracy-Morph, with a stronger BabyLM backbone; that experiment would answer a different question from the one isolated in this paper.

9 Ethics Statement

This work studies corpus selection for small language model training. The approach may affect which linguistic varieties and domains are represented in the selected corpus. We therefore report corpus statistics and encourage analysis of distributional changes introduced by selection.

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To be completed.

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A Implementation Details

The code is organized in `./babylm-hhm`. Core masking classes are in `src/masking/`. Configurations cover the random baseline, HHM, entity-only, error-only, randomized sketch control, frequency-only control, CMS-Morph, Accuracy-Morph, and Entity-Accuracy-Morph. The official BabyLM evaluation repository is available locally.

B Reproducibility Commands

Tests are run with the local test runner. The three-seed main comparison uses the experiment runner with the main suite and seeds 1–3. The preliminary ablations use the same runner with the ablations suite. The adaptive pilots use the training entry point with the CMS-Morph, Accuracy-Morph, and Entity-Accuracy-Morph config files. All full training runs were executed with CUDA visible.